## Problem 16.3

A 5-year annuity due has periodic cash flows of \$100 each year. Find the accumulated value of this annuity if the effective annual interest rate is 8%.

## Solution

Since this is an annuity due, payments are made at the beginning of each year. The accumulated value is

$$AV = 100\ddot{s}_{\overline{5}|}.$$

For an annuity due,

$$\ddot{s}_{\overline{n}|} = (1+i)s_{\overline{n}|}.$$

Using $i = 0.08$,

$$\ddot{s}_{\overline{5}|} = (1.08)\left(\frac{(1.08)^5 - 1}{0.08}\right).$$

Thus,

$$\begin{aligned} AV &= 100(1.08)\left(\frac{(1.08)^5 - 1}{0.08}\right) \\ &\approx 633.5929. \end{aligned}$$

Hence, the accumulated value is

$$\boxed{\$633.59}.$$